E80 Autonomous Robot for Depth-Based Sensing of Light Intensity, pH, and Temperature

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E80: Experimental Engineering, SP 26

Problem Statement

The water quality at Dana Point has been in question for years after being in close proximity with a sewage pipe creating a potentially dangerous environment for the organisms living there. Environmental water quality varies with depth, but collecting depth-resolved data is difficult without complex equipment. There is a need for an autonomous underwater vehicle (AUV) that can measure key environmental variables such as light intensity, temperature, and pH as a function of depth in real-world conditions.

Impact

- Data of pH, temperature, and light allows for easy identification of areas with bad water quality
- Strong data can create priority for change and action in water quality at Dana Point
- Detecting dangerous areas quickly can help protect local organisms from being in unsafe conditions

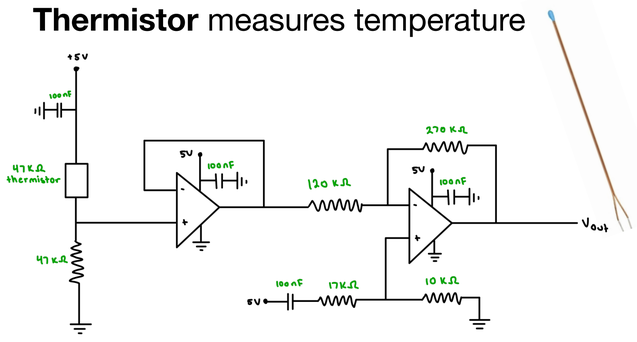
Future Work

- Improvement of pH sensor to refine data and revise interpretations
- Research on liveable conditions for marine life in bodies of water similar to Dana Point
- Test local organism health at Dana Point to see expected vs actual results

Autonomous Underwater Vehicle (AUV) Features

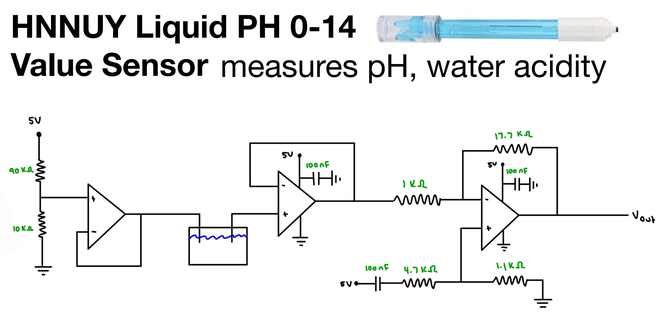
Electrical Schematic & Sensors

Murata NXFT15WB473FA2B150 NTC Thermistor measures temperature



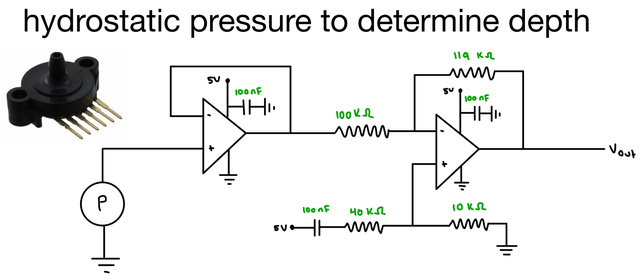
Expected input of sensor: 47k - 95.5k
Expected output of circuit: 0.5 - 2.7 V

HNNUY Liquid PH 0-14 Value Sensor measures pH, water acidity



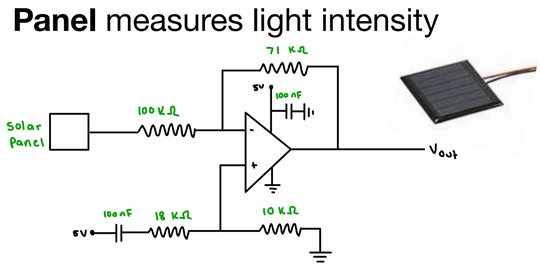
Expected input of sensor: 0.779 - 0.9 V
Expected output of circuit: 0.5 - 2.7 V

MPX5700ASX Pressure Sensor measures hydrostatic pressure to determine depth



Expected input of sensor: 0.851 - 1.040 V
Expected output of circuit: 0.5 - 2.7 V

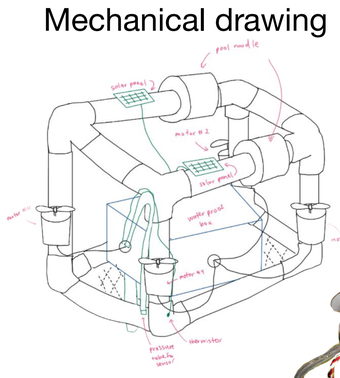
JamesCo ZW-5050-3V Solar Panel measures light intensity



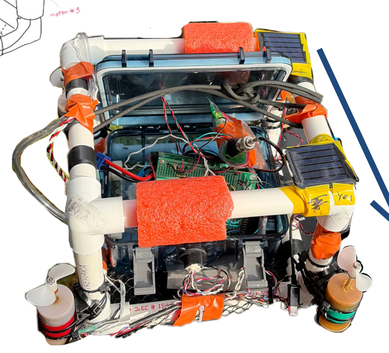
Expected input of sensor: 0.5 - 3.6 V
Expected output of circuit: 0.5 - 2.7 V

Mechanical Design & Integration

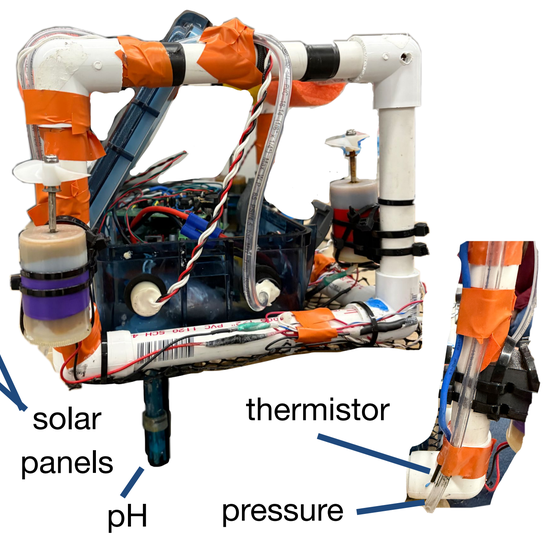
Mechanical drawing



Aerial view



Side view of various sensors



Design goal: stable vertical diving while maintaining consistent sensor orientation and waterproofness during deployment

The pressure sensor and thermistor were placed downward to measure hydrostatic pressure and temperature in flowing water, the solar panels were oriented upward to catch incident light, and pH probe was isolated due to output impedance and oriented downward to measure water acidity.

Experimental Data and Results

Sensor Calibration

Pressure	Thermistor	pH	Solar Panel
Calibrated in Roberts Pool (0 to 3 m), linear fit	Calibrated with water baths (10 to 25 C), linear fit	Calibrated with buffer solutions (pH 7 to 9), linear fit	Calibration using iPhone flashlight (lumens), quadratic fit
h = -1.471 V + 3.9223	T = -11.854 V + 32.433	pH = 0.912 V + 6.538	I = 88.58 V^2 - 427.97 V + 512.27
Se = 0.0462 m	Se = 4.131 C	Se = 0.0899 pH	R^2 = 0.9803

Ground Truth Measurements

Atmospheric pressure	101.25 kPa
Surface ocean temperature	18.7 C
Ocean pH	~7.4
UV	2-3

Pressure matched expected depth trends. Temperature showed moderate cooling with depth and light intensity decreased with depth. pH trends observed but not quantitatively validated with depth.

Results & Conclusions

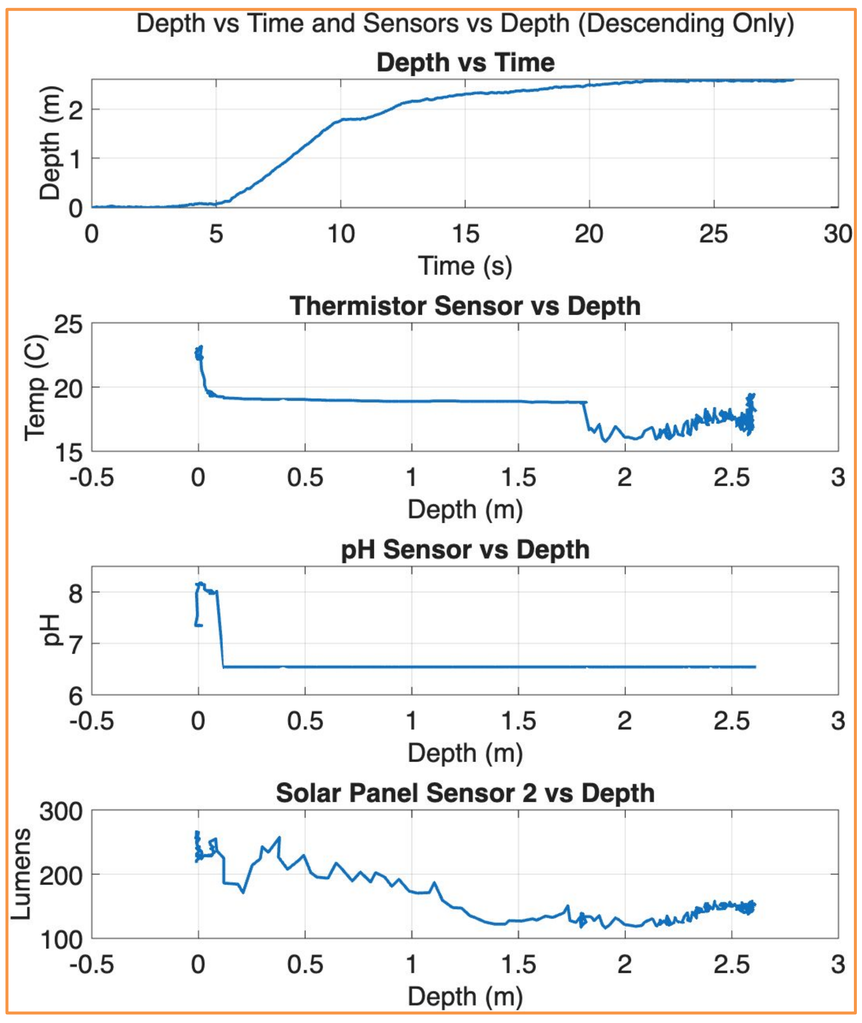
The AUV successfully measured depth-dependent light and temperature trends, but pH sensing needs to be refined for reliable environmental monitoring.

Pressure Sensor (Depth)
- Depth increases with time, reached ~2.6 m maximum depth

Thermistor Sensor (Temperature)
- Thermistor settles to then decreases to 16°C as robot dove
- 14.4% decrease in temperature from ground truth measurement 18.7°C at surface

pH Sensor
- Sensor railed out at deployment
- Ground truth of 7.4 pH, forensic result in lab showed around 6.5 pH with a tolerance of ±1 pH
- Error attributed to glass electrode instability in salt water

Solar Panel (Light Intensity in Lumens)
- Panels show a 52% decrease in light intensity with depth



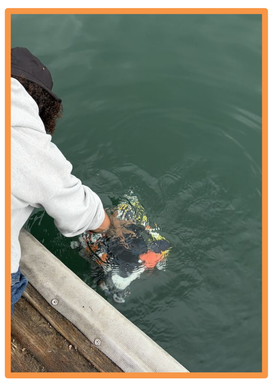
Deployment Details

Location(s): Dock at Dana Point, California

Testing Protocol: Before deployment, we made sure that all of the outputs were transferring data to the analog pins by connecting the Teensy 4.0 to the Serial monitor in Arduino IDE. Deployment safety was addressed by waterproofing the electronics box and strategic wire management to prevent electrical shorting, using a delayed motor start to prevent injuries, and recovering the robot immediately.

If waterproofing fails, the tether is used to immediately retrieve the robot to prevent permanent loss and limit damage.





Acknowledgements

We would like to thank Xavier Walter, Lynn Kim, Jacob Staimpel, Georgia Tai, Tracy Han, Joshua Ikehara, Prof. Yang and all the other E80 professors and TAs for helping us through our E80 journey. The HMC stockroom provided key sensing components, while Machine Shop and Makerspace provided waterproofing materials.